

KEONWOO OH

✉ koh3@umd.edu  github.com/keonoh96 📍 Maryland, United States

EDUCATION

University of Maryland

Aug 2021 - Present

Ph.D. in Computer Science, *Advisor:* Amol Deshpande

GPA: 3.8/4.0

Swarthmore College

Aug 2015 - May 2021

B.A. in Computer Science and Mathematics

GPA: 3.9/4.0

RESEARCH INTERESTS

Metadata management and **concurrency control mechanism** in large-scale data lakehouse systems. **Data formats** for efficient data asset storage in data and machine learning pipelines. Broader topics in **database system design**.

PUBLICATIONS

Keonwoo Oh, Pooja Nilangekar, and Amol Deshpande. *TreeCat: Standalone Catalog Engine for Large Data Systems*. Proc. VLDB Endow. 18, 11 (July 2025), 4323-4336.

PROJECTS

Raven

Implemented a file-level metadata system for data lakehouse architectures with a specialized focus on garbage collection.

TreeCat (<https://github.com/umddb/treecat>)

Implemented a unified, standalone catalog engine for large data lake systems. Published in Proc. VLDB 18, 11 (July 2025).

WORK EXPERIENCE

Dolby Laboratories

May 2026 - Aug 2026

PhD Research Intern

Atlanta, GA

- Evaluated and benchmarked tabular storage formats for multimodal data management in ML pipelines.

Dolby Laboratories

May 2025 - Aug 2025

PhD Research Intern

Atlanta, GA

- Designed and implemented a file-level metadata system for data lakehouse architectures.

Vertica

May 2022 - Aug 2022

Systems Software Engineering Intern

Cambridge, MA

- Integrated the internal profiling tool into the HTTPS service via a REST API.

TEACHING EXPERIENCE

Teaching Assistant, CMSC416: Introduction to Parallel Computing, UMD

Spring 2023, Spring 2024, Fall 2024

Teaching Assistant, CMSC714: High Performance Computing Systems, UMD

Fall 2023

Teaching Assistant, CMSC424: Database Design, UMD

Fall 2022

Teaching Assistant, CMSC250: Discrete Structures, UMD

Spring 2022

Teaching Assistant, CMSC216: Introduction to Computer Systems, UMD

Fall 2021

SKILLS

Programming Languages: C++, C, Scala, Python, Java, Rust, OCaml

Systems: PostgreSQL, MySQL, Apache Spark, Apache Iceberg, Delta Lake, DuckDB, RocksDB, gRPC